

DAY 1 — IP ADDRESSING BASICS

What is an IP Address?

An IP Address (Internet Protocol Address) is a unique numerical address assigned to devices connected to a network.

It helps devices communicate with each other over the internet or local network.

Example:

192.168.1.10

Why IP Addresses are Important

IP addresses help:

- Identify devices
- Send and receive data
- Track communication
- Access websites
- Monitor security events

In IAM (Identity and Access Management), IP addresses are used for:

- Login monitoring
- Conditional access
- Detecting suspicious logins
- Geo-location security
- Access restrictions

Types of IP Addresses

1. Public IP Address

A public IP is used on the internet.

Example:

49.204.x.x

Assigned by Internet Service Provider (ISP).

2. Private IP Address

Used inside local/private networks.

Private IP ranges:

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

Example:
192.168.1.5

IPv4 Basics

IPv4 consists of:

- 4 octets
- Numbers between 0–255

Example:
192.168.1.1

Important Networking Commands

ipconfig

Shows IP configuration.

ping

Checks connectivity.

Example:
ping google.com

nslookup

Checks DNS resolution.

Example:
nslookup google.com

Difference Between Static and Dynamic IP

Static IP

- Fixed IP
- Does not change

Dynamic IP

- Automatically assigned
- Changes periodically

Role of IP Address in IAM

IAM systems use IP addresses for:

- Authentication monitoring
- Login tracking
- Conditional access policies

- Zero Trust security
- Suspicious login detection

Key Learnings

- Every device needs an IP address
- IP addresses help communication
- Private IPs are used inside organizations
- Networking basics are important for IAM Analysts